

3	(a) mass / volume (ratio idea essential)	B1	[1]
	(b) (i) mass = $Ah\rho$	B1	[1]
	(ii) pressure = force/area	B1	
	weight (of liquid)/force (on base) = $Ah\rho g$	B1	
	pressure = $h\rho g$	A0	[2]
	(c) (i) ratio = 1600 or 1600:1	A1	[1]
	(ii) ratio = $\sqrt[3]{1600}$	C1	
	= 11.7 (allow 12)	A1	[2]

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(d) (i) density of solids and liquids are (about) equal B1 [1]

(ii) strong forces: fixed volume B1
 rigid forces: retains shape / does not flow / little deformation B1 [2]
(allow 1 mark for fixed volume, fixed shape)

4 (a) (i) (change in) potential energy = mgh C1